

USER CENTERED DESIGN



Conceptual Design

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Technology

- The application of scientific knowledge for practical purposes



Human Capacity

- Human knowledge, habits, social and personality attributes including creativity, embodied together to perform the ability to make development and progress possible.



Technology VS. HUMAN CAPACITY

- relation between the level of human capital and total factor productivity growth or technological change is important; the emphasis here is on the link between human capital and disembodied knowledge as manifested in technology



Technology VS. HUMAN CAPACITY

- Technical progress can be understood as the process of new products development, and understanding how knowledge and skills contribute to this process can shed light on the introduction of new technologies



Technology VS. HUMAN CAPACITY

- The idea draws on earlier insights on the link between R&D, innovation and market value in Schumpeter (1934) and Griliches (1981) and is central in the first generation of endogenous growth models where human capital is the engine of innovation and sustainable growth (Romer 1990; Aghion and Howitt 1998).



Technology VS. HUMAN CAPACITY

- Here, human capital is a key determinant of absorptive capacity since it enables workers to understand and assimilate new technology; a particular formulation of the convergence process whereby less developed economies catch-up with the developed world



Technology VS. HUMAN CAPACITY

- The idea originates in Nelson and Phelps (1966) who assessed education to be a catalyst in the diffusion of new technologies. Their model rests on two key assumptions: the further an economy is from the technology frontier, the stronger is the incentive to exploit externalities; and the bigger the human capital the greater is the capability to learn and adopt the new technology.



Technology VS. HUMAN CAPACITY

- Benhabib and Spiegel (1994) integrate the two ideas in a generalised model of human capital that aims to explain both innovation and technology diffusion.



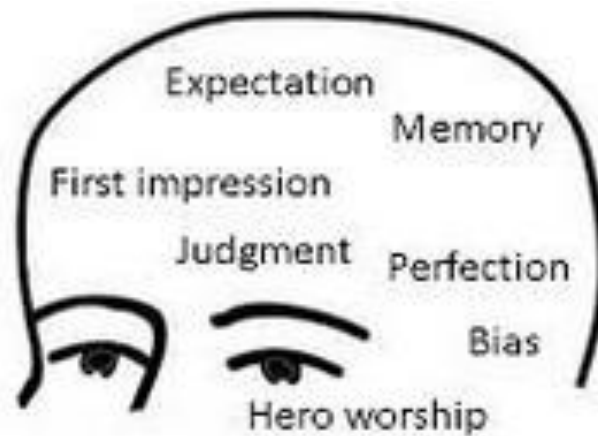
Technology VS. HUMAN CAPACITY

- They build on the intuition that the two views of human capital are complementary rather than competing, for they explain different stages of economic development; i.e., nations closer to the technology frontier have accumulated high levels of human capital that could support innovation while countries far from the frontier focus on technology diffusion



Mental models

- A mental model is an explanation of someone's thought process about how something works in the real world.



Mental models

- It is a representation of the surrounding world, the relationships between its various parts and a person's intuitive perception about his or her own acts and their consequences.



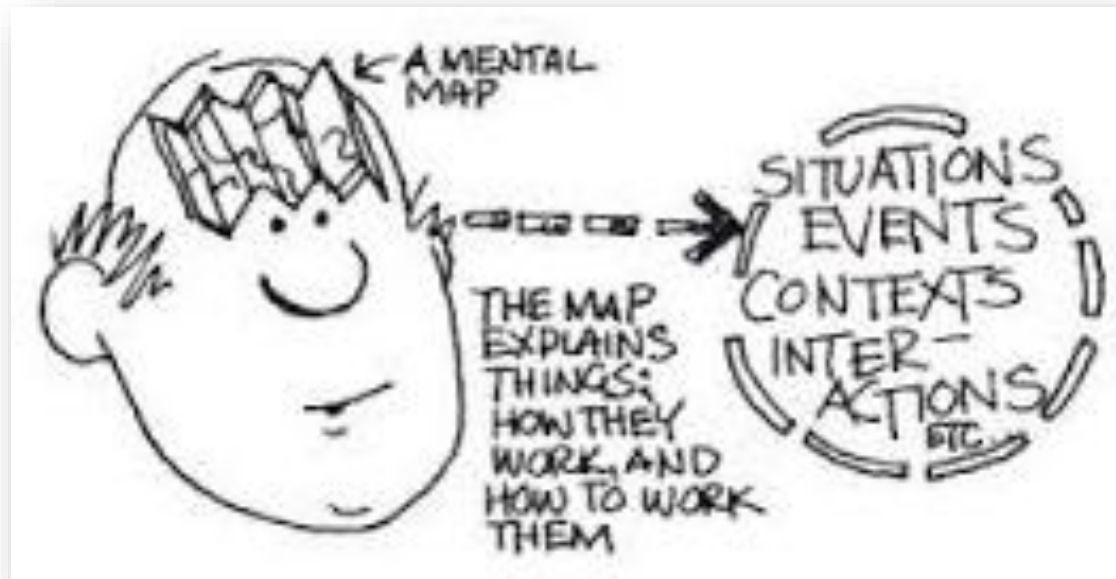
Mental models

- Mental models can help shape behavior and set an approach to solving problems (akin to a personal algorithm) and doing tasks.



Mental models

- A mental model is a kind of internal symbol or representation of external reality, hypothesized to play a major role in cognition, reasoning and decision-making.



Mental models

- Kenneth Craik suggested in 1943 that the mind constructs "small-scale models" of reality that it uses to anticipate events.



Mental models

- Jay Wright Forrester defined general mental models as:
 - The image of the world around us, which we carry in our head, is just a model. Nobody in his head imagines all the world, government or country. He has only selected concepts, and relationships between them, and uses those to represent the real system.



Conceptualizing model

- User Requirements => Conceptual Model
- “a description of the proposed system in terms of a set of integrated ideas and concepts about what it should do, behave, and look like, that will be understandable by the users in the manner intended”
- High-level compared to physical design



Guiding principles

- Discuss and get different perspectives
- Prototyping
- Iterate, iterate, iterate!
- Consider many alternatives
- “To get a good idea, get lots of ideas”
- Empathize with the user



Developing a Conceptual Model

- Three perspectives
 - Interaction Mode
 - How the user invokes actions when interacting with the device



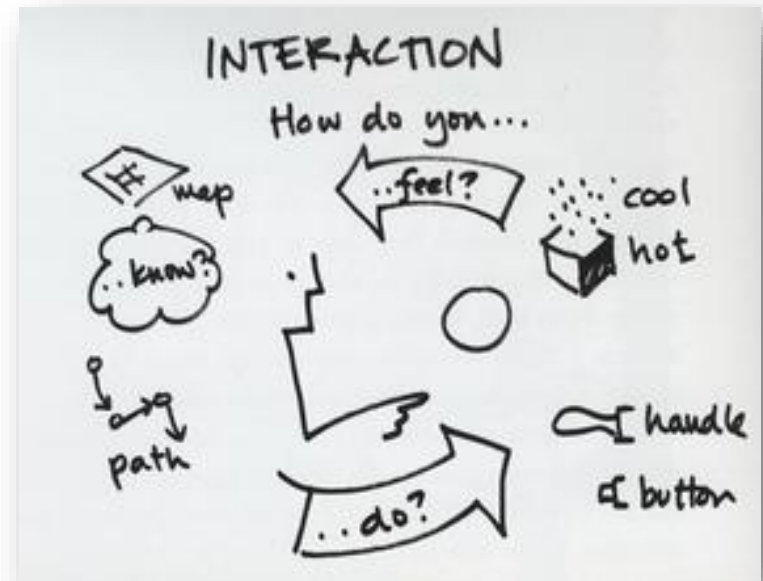
Developing a Conceptual Model

- Three perspectives
 - Interface Metaphor
 - Combining familiar knowledge with new knowledge in a way that helps the user understand the system



Developing a Conceptual Model

- Three perspectives
 - Interaction Paradigm
 - Particular way of thinking about interaction design



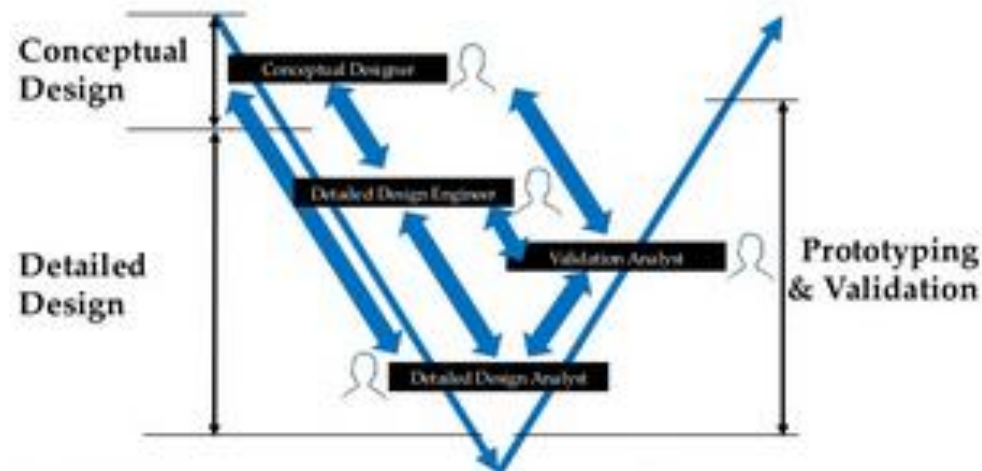
Expanding the conceptual model

- How are the functions related to each other?
 - Sequential or parallel?
 - Categorisations, e.g. all actions related to telephone memory storage



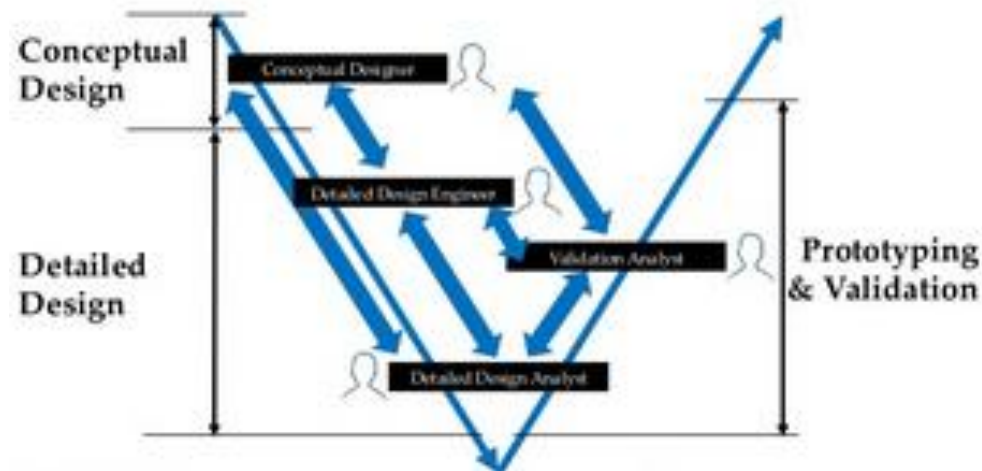
Conceptual design: from requirements to design

- Transform user requirements/needs into a conceptual model
- “a description of the proposed system in terms of a set of integrated ideas and concepts about what it should do, behave and look like, that will be understandable by the users in the manner intended”



Conceptual design: from requirements to design

- Don't move to a solution too quickly. Iterate, iterate, iterate
- Consider alternatives: prototyping helps



Expanding the conceptual model

- What functions will the product perform?
 - What will the product do and what will the human do (task allocation)?



Using scenarios in conceptual design

- Express proposed or imagined situations
- Used throughout design in various ways
 - scripts for user evaluation of prototypes
 - concrete examples of tasks
 - as a means of co-operation across professional boundaries
- Plus and minus scenarios to explore extreme cases



Expanding the conceptual model

- What information needs to be available?
 - What data is required to perform the task?
 - How is this data to be transformed by the system?



Is there a suitable metaphor?

- Interface metaphors combine familiar knowledge with new knowledge in a way that will help the user understand the product.
- Three steps: understand functionality, identify potential problem areas, generate metaphors



Thought



Idea



Goodidea



Great idea



Bluesky idea



© Present Recycled idea

Is there a suitable metaphor?

- Evaluate metaphors:
 - How much structure does it provide?
 - How much is relevant to the problem?
 - Is it easy to represent?
 - Will the audience understand it?
 - How extensible is it?



Considering interaction types

- Which interaction type?
 - How the user invokes actions
 - Instructing, conversing, manipulating or exploring
- Do different interface types provide insight?
 - WIMP, shareable, augmented reality, etc



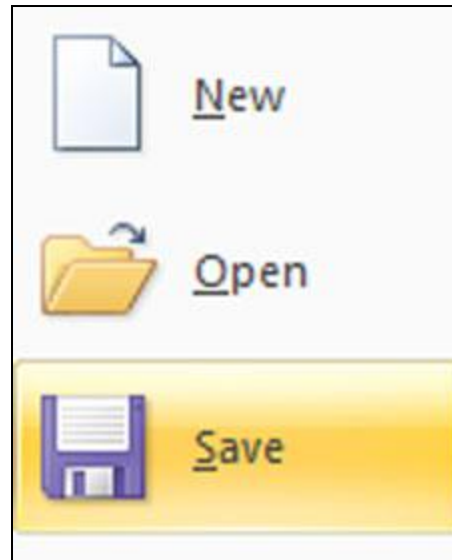
Interaction Modes

- Activity-based
 - Instructing
 - Conversing
 - Manipulating & Navigating
 - Exploring & Browsing
- Object-based

Interaction Modes

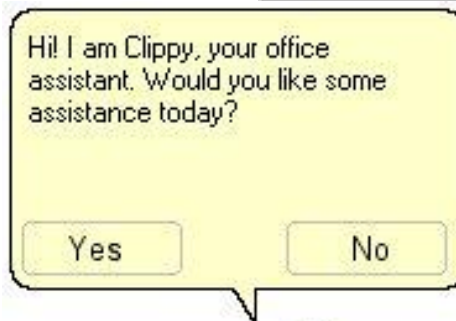
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>ping ics.uci.edu
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Interaction Modes

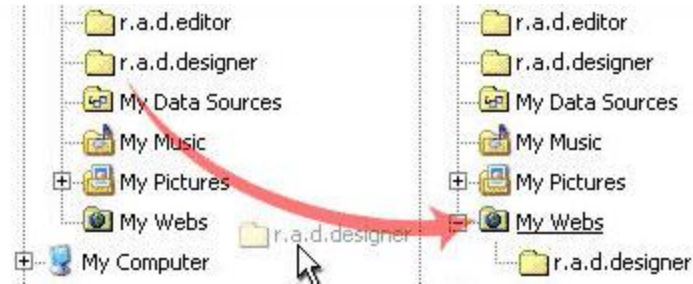
- Activity-based
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- Object-based



Press 1 for customer service.
Press 3 for store hours.

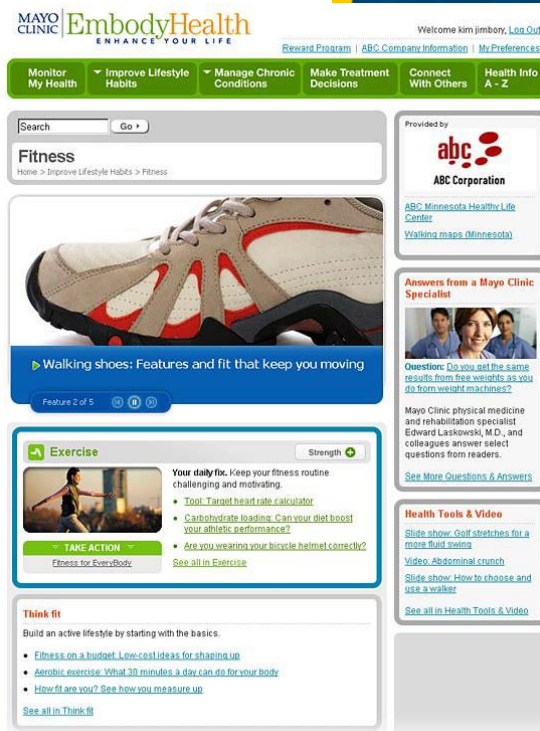
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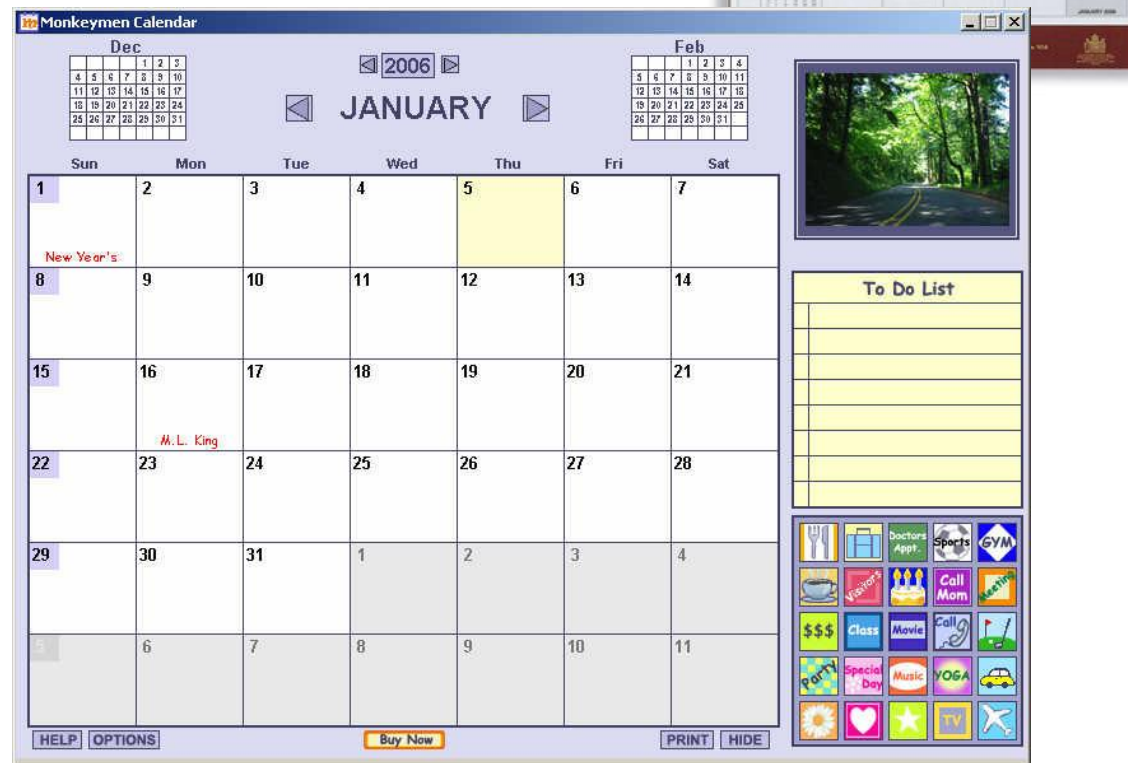
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Interface Paradigms

- Desktop
- Ubiquitous
- Pervasive
- Wearable
- ...



- Consider user tasks & environmental requirements

Design Principles (Norman, 1990)

- Donald Norman, in his book *The Design of Everyday Things*, introduced several basic user interface design principles and concepts that are now considered critical for understanding why some designs are more usable and learnable than others:
 - **Visibility**
 - **Feedback**
 - **Constraints**
 - **Mapping**
 - **Consistency**
 - **Affordance**



Design Principles (Norman, 1990)

- **Visibility** – The more visible functions are, the more likely users will be able to know what to do next. Incontrast, when functions are "out of sight," it makes them more difficult to find and know how to use.



Design Principles (Norman, 1990)

- **Feedback** – Feedback is about sending back information about what action has been done and what has been accomplished, allowing the person to continue with the activity. Various kinds of feedback are available for interaction design-audio, tactile, verbal, and combinations of these.



Design Principles (Norman, 1990)

- **Constraints** – The design concept of constraining refers to determining ways of restricting the kind of user interaction that can take place at a given moment. There are various ways this can be achieved.



Design Principles (Norman, 1990)

- **Mapping** – This refers to the relationship between controls and their effects in the world. Nearly all artifacts need some kind of mapping between controls and effects, whether it is a flashlight, car, power plant, or cockpit. An example of a good mapping between control and effect is the up and down arrows used to represent the up and down movement of the cursor, respectively, on a computer keyboard.



Design Principles (Norman, 1990)

- **Consistency** – This refers to designing interfaces to have similar operations and use similar elements for achieving similar tasks. In particular, a consistent interface is one that follows rules, such as using the same operation to select all objects. For example, a consistent operation is using the same input action to highlight any graphical object at the interface, such as always clicking the left mouse button. Inconsistent interfaces, on the other hand, allow exceptions to a rule.



Design Principles (Norman, 1990)

- **Affordance** – is a term used to refer to an attribute of an object that allows people to know how to use it. For example, a mouse button invites pushing (in so doing acting clicking) by the way it is physically constrained in its plastic shell. At a very simple level, to afford means "to give a clue" (Norman, 1988). When the affordances of a physical object are perceptually obvious it is easy to know how to interact with it.



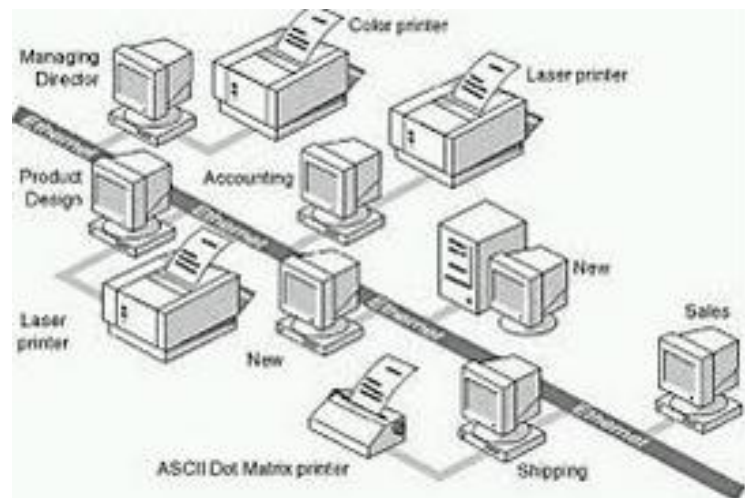
Physical Design

- Conceptual design abstractly describes system's intended behavior
 - ATM should authenticate user ID and allow user to withdraw cash on command



Physical Design

- Physical design addresses specific, concrete layout and design issues
 - ATM should have a card reader slot, a 10 digit keypad for users to enter a PIN number, a touch screen monitor with a menu of command options and a mechanical reel for dispensing increments of \$20 bills



Physical Design Guidelines

- Principles
 - Abstract statements open to interpretation
 - e.g., Maintain consistency, keep designs simple, support user recognition vs. recall, etc.
 - Mostly same principles reviewed in first half of the quarter



Physical Design Guidelines

- Rules
 - Specific statements, with no interpretations
 - e.g., Menus should have no more than 8 options



Physical Design Guidelines

- Style Guides and Standards
 - Collections of principles and rules to achieve consistency across applications. Good for corporate identity and consistency.
 - e.g., Windows or Mac style guides: File menu first, Help menu last, etc. Standard icon for save, cut, copy, paste, etc.
 - Useful to adopt *ad hoc* standards to meet user expectations

Specific Widget Guidelines

- Menu Design
 - Common functions should be easiest to reach
 - >8 options is too much, grouping to organize
 - Opposite and dangerous operators should be physically separated to avoid accidents



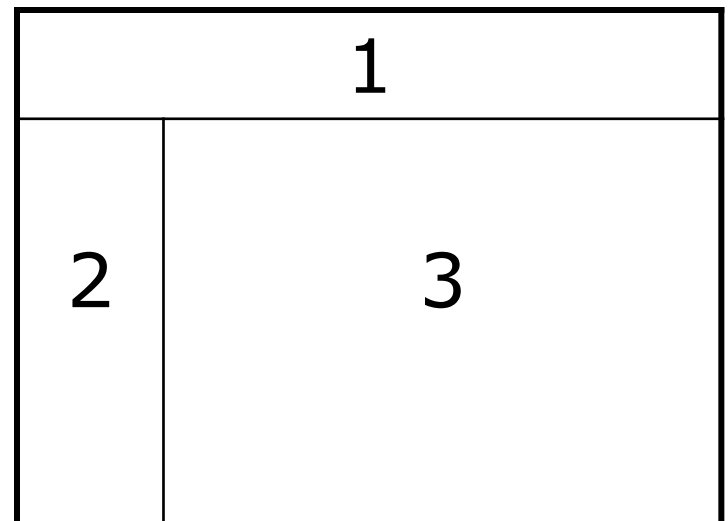
Specific Widget Guidelines

- Icon Design
 - Takes time to develop a good one
 - Immediately recognizable (small and simple)
 - Easily distinguishable from others
 - More important than being very descriptive



Specific Widget Guidelines

- Web Design Specifics
 - Unique with *hyperlink nature, short user attention span, download lag time*
 - Key questions web pages should answer within 3 seconds of scanning
 1. Where am I?
 2. Where can I go?
 3. What's here?



Specific Widget Guidelines

- Multi-Screen Designs
 - Task flow with multiple steps
 - Start with one page per “atomic” step, but balance against too many trivial pages
 - Consider frames or overlapping windows to easily move between non-linear task flow
 - Pertinent info should be available across all steps

Design cycles of interactive system

- Evaluating

- Evaluation is the process of determining the usability and acceptability of the product or design that is measured in terms of a variety of criteria including the number errors users make using it, how appealing it is, how well it matches the requirements, and so on.



Evaluation

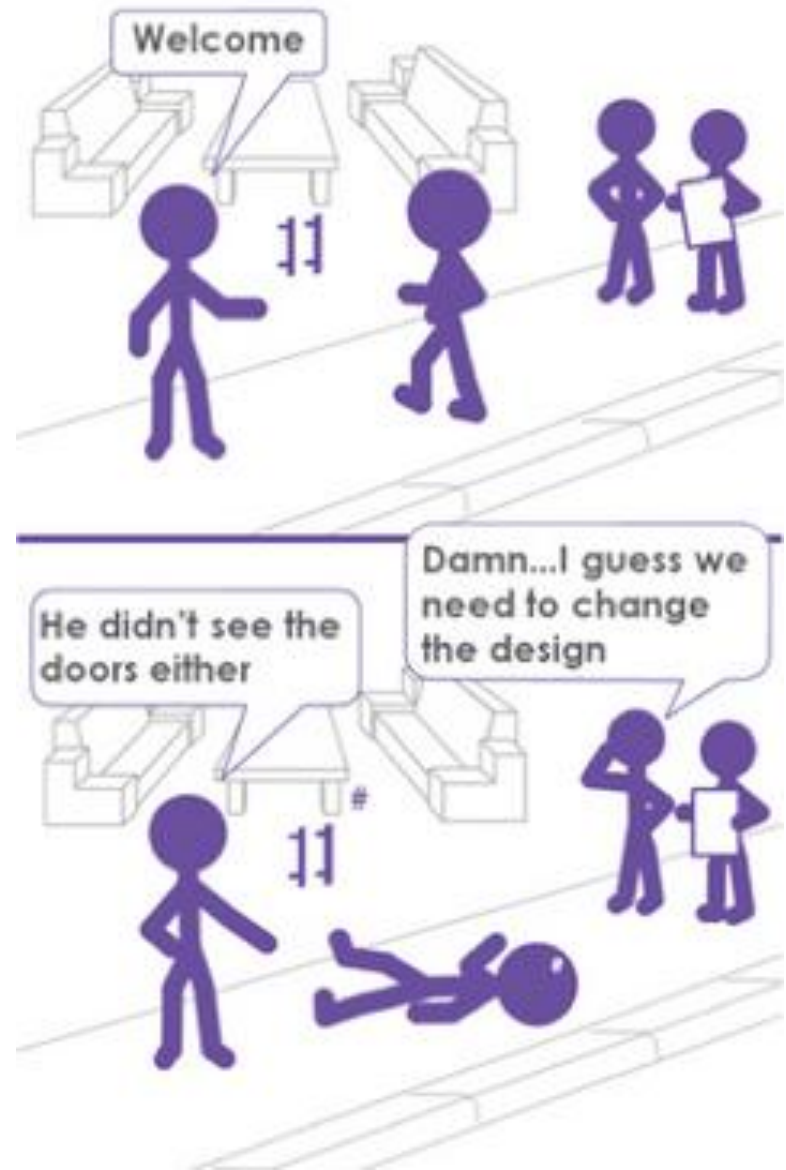
Iterative design & evaluation is a continuous process that

examines:

- Why: to check users' requirements and that users can use the product and they like it.
- What: a conceptual model, early prototypes of a new system and later, more complete prototypes.
- Where: in natural and laboratory settings.
- When: throughout design; finished products can be evaluated to collect information to inform new products.

Evaluation

- Evaluation is integral to the design process.



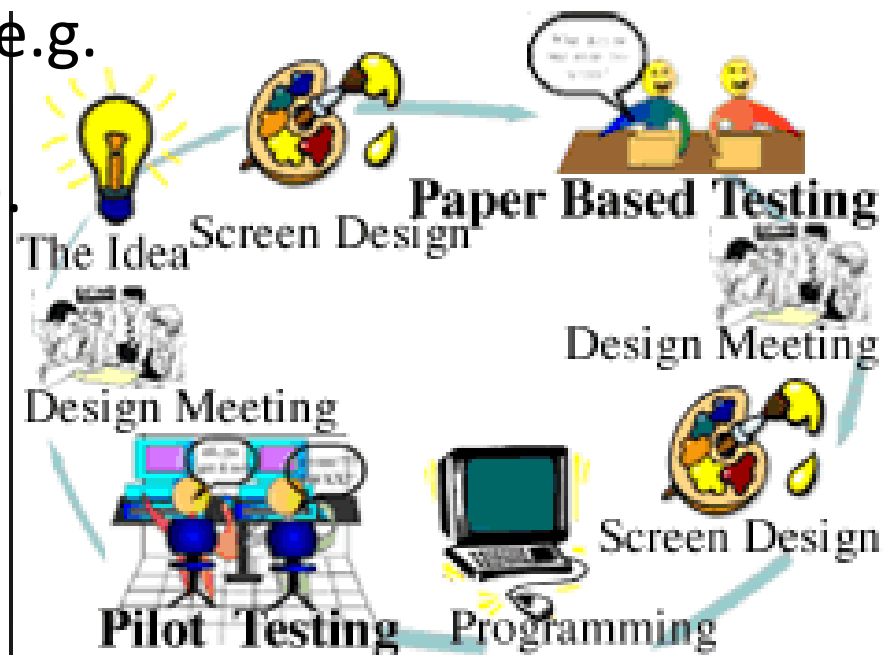
Evaluation

- Evaluators collect information about users' or potential users' experiences when interacting with a prototype, a computer system, a component of a computer system, an application, or a design artefact such as a screen sketch.



Evaluation

- They do this in order to improve its design. Evaluation focuses on both the usability of the system (e.g. how it is to learn and to use) and on the users' experience when interacting with the system (e.g. how satisfying, enjoyable, or motivating the interaction is).



Bruce Tognazzini tells you why you need to evaluate

- “Iterative design, with its repeating cycle of design and testing, is the only validated methodology in existence that will consistently produce successful results. If you don’t have user-testing as an integral part of your design process you are going to throw buckets of money down the drain.”

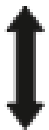
Types of evaluation

- Controlled settings involving users, eg usability testing & experiments in laboratories and living labs.
- Natural settings involving users, eg field studies to see how the product is used in the real world.
- Any settings not involving users, eg consultants critique; to predict, analyze & model aspects of the interface analytics.

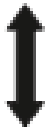
Usability testing & field studies can compliment



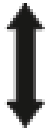
Field study to evaluate initial design ideas and get early feedback



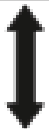
Make some design changes



Usability test to check specific design features



Field study to see what happens when used in natural environment



Make some final design changes

Evaluation case studies

- Experiment to investigate a computer game
- In the wild field study of skiers
- Crowdsourcing

Evaluation methods

Method	Controlled settings	Natural settings	Without users
Observing	X	X	
Asking users	X	X	
Asking experts		X	X
Testing	X		
Modeling			X

The language of evaluation

- Analytics
- Analytical evaluation
- Controlled experiment
- Expert review
- Field study
- Formative evaluation
- Heuristic evaluation
- In the wild evaluation
- Living laboratory
- Predictive evaluation
- Summative evaluation
- Usability laboratory
- User studies
- Usability testing
- Users or participants

Key points

- Evaluation & design are closely integrated in user-centered design.
- Some of the same techniques are used in evaluation as for establishing requirements but they are used differently (e.g. observation interviews & questionnaires).

Key points

- Three types of evaluation: laboratory based with users, in the field with users, studies that do not involve users
- The main methods are: observing, asking users, asking experts, user testing, inspection, and modeling users' task performance, analytics.
- Dealing with constraints is an important skill for evaluators to develop.

End of Lecture